Using Generative AI Models to Support Cybersecurity Analysts

Abstract

One of the tasks of security analysts is to detect security vulnerabilities and ongoing attacks. There is already a large number of software tools that can help to collect security-relevant data, such as event logs, security settings, application manifests, and even the (decompiled) source code of potentially malicious applications. The analyst must study these data, evaluate them, and properly identify and classify suspicious activities and applications. Fast advances in the area of Artificial Intelligence have produced large language models that can perform a variety of tasks, including generating text summaries and reports. In this article, we study the potential black-box use of LLM chatbots as a support tool for security analysts. We provide two case studies: the first is concerned with the identification of vulnerabilities in Android applications, and the second one is concerned with the analysis of security logs. We show how LLM chatbots can help security analysts in their work, but point out specific limitations and security concerns related to this approach.

Keywords: LLM; Cybersecurity; vulnerability assessment

🔍 What the paper is about:

The paper explores **how Large Language Models (LLMs)** — like ChatGPT — can assist **Cybersecurity analysts** in identifying **vulnerabilities and attacks**.

🛠️ Current Situation:

Security analysts already use many tools that collect data such as:

- Event logs
- Security settings
- Application details (e.g., source code of apps)

However, analysts still need to manually **analyze**, **evaluate**, and **classify** this information to detect threats.

🤖 New Idea:

With the rise of **AI and LLMs**, tools like chatbots can now:

- Summarize data

- Generate reports
- Provide insights from raw security data

This paper investigates whether these LLMs can be used **as "black-box" assistants** — meaning, the analyst gives input and gets results without needing to know how the model works internally.

📊 **Case Studies in the Paper:**

1. **Android App Vulnerability Detection**
    - Uses LLMs to help find flaws in mobile apps.
2. **Security Log Analysis**
    - Uses LLMs to assist in making sense of event logs.

⚠️ **Findings and Concerns:**

- **Pros**: LLMs can **help analysts work faster** and more effectively.
- **Cons**: There are **limitations** (e.g., model hallucination, lack of explainability) and **security risks** (e.g., leaking sensitive data, incorrect assessments).

🧩 **Keywords Explained:**

- **LLM**: Large Language Models like ChatGPT
- **Cybersecurity**: Protecting systems from digital attacks
- **Vulnerability Assessment**: Identifying weaknesses in software/systems

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Visual summary :

🛡️ **Using Generative AI to Help Cybersecurity Analysts**

🔍 Problem

Cybersecurity analysts deal with **huge amounts of data**
(e.g., event logs, app code, settings)
➤ Manual analysis is time-consuming and complex

🤖 Solution Explored

Use **Large Language Models (LLMs)** like ChatGPT as **assistants** to:

- Summarize security data
- Help detect vulnerabilities
- Interpret logs

🧪 Case Studies

Case Study	Description
📱 *Android Vulnerability Detection*	LLMs analyze Android apps for potential flaws
📋 *Security Log Analysis*	LLMs interpret logs to find suspicious behavior

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✅ Benefits

- Faster analysis
- Better summaries
- Reduces manual workload

⚠️ Challenges

- ❗ LLMs may hallucinate or give wrong answers
- 🔒 Risk of leaking sensitive data
- 🧠 Lack of explainability (black-box behavior)

💡 Conclusion

LLMs can be **valuable tools**, but **must be used with caution** due to risks.

USING GENERATIVE AI TO HELP CYBERSECURITY ANALYSTS

PROBLEM

Cybersecurity analysts deal with **HUGE AMOUNTS OF DATA** (e.g., event logs, app code, settings)

➔ Manual analysis is time-consuming and complex

SOLUTION EXPLORED

Use LARGE LANGUAGE MODELS (LLMs) like ChatGPT as assistants to:

- Summarize security data
- Help detect vulnerabilities
- Interpret logs

Case Study	Description
ANDROID VULNERABILITY DETECTION	LLMs analyze Android apps for potential flaws
SECURITY LOG ANALYSIS	LLMs interpret logs to find suspicious behavior

BENEFITS

- Faster analysis
- Better summaries
- Reduces manual workload

CHALLENGES

⚠ LLMs may hallucinate or give wrong answers

🔒 Risk of leaking sensitive (black-box behavior)

🧠 How LLMs Are Used in Cybersecurity Tasks

🔧 1. **Data Summarization**

- LLMs can **digest long logs or reports** and provide concise summaries.
- Useful for daily Security Operation Center (SOC) briefings.
- Reduces human fatigue from reading repetitive entries.

🔎 2. **Threat Classification**

- Analysts can input unusual log entries or code snippets.
- The LLM suggests likely threat categories (e.g., malware, phishing, misconfig.)
- Speeds up **triage decisions**.

📂 3. **Reverse Engineering Aid**

- De-compiling apps produces hard-to-read source code.
- LLMs can help **explain the purpose** of suspicious code blocks.
- Especially helpful for Android APK analysis in malware research.

🛠️ 4. **Incident Report Drafting**

- After an incident, LLMs assist in writing reports:
    - What happened?
    - What was affected?
    - What actions were taken?
- Ensures quicker documentation.

🚨 Limitations & Mitigation Strategies

Limitation	Description	Suggested Mitigation
❌ **Hallucination**	LLMs may invent incorrect facts	Use verification steps with known databases/tools

Limitation	Description	Suggested Mitigation
🔐 **Data Leakage**	Inputting raw data may expose secrets	Run LLMs **locally** or on **secured air-gapped systems**
📦 **Black-box behavior**	Models don't explain why they respond a certain way	Use LLMs as **support**, not final authority

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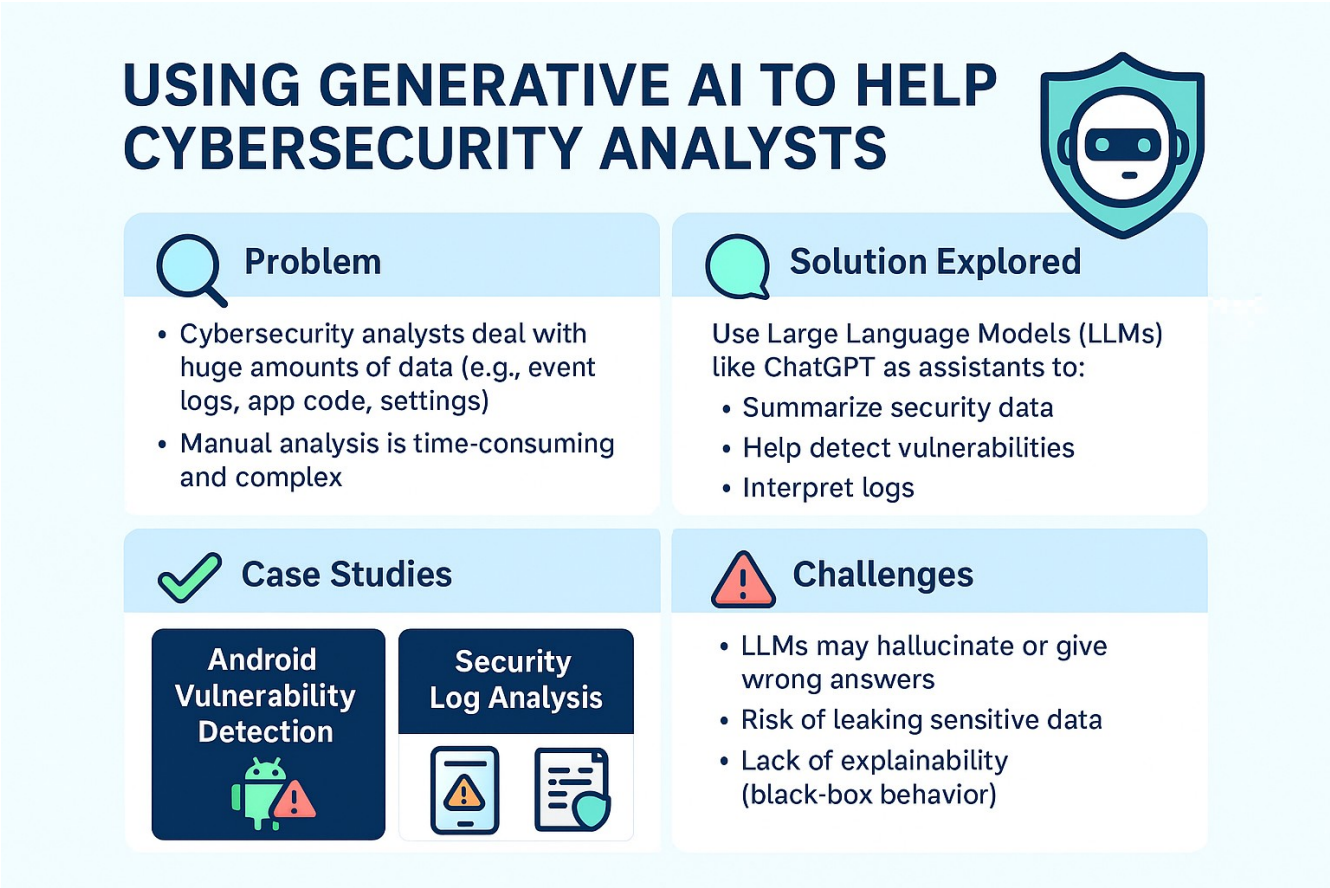
🧪 Future Scope & Research Areas

1. **Fine-tuning LLMs on Cybersecurity datasets**
    - Improves reliability and accuracy.
2. **LLMs integrated with SIEM tools**
    - E.g., Splunk + GPT for interactive queries.
3. **Chain-of-Thought prompting**
    - Asking LLMs to "think step-by-step" before answering.

📘 Real-World Use Cases

- **Google's Chronicle Security**: Integrates AI to help search and contextualize threat data.
- **Microsoft Security Copilot**: Uses GPT-4 for assisting enterprise defenders.
- **Cybersecurity startups**: Developing custom AI co-pilots for security teams.



Explanation of the **info-graphic** : **"Using Generative AI to Help Cybersecurity Analysts"**

🧩 1. **Title and Theme**

The title at the top clearly sets the context:

> It's about using **Large Language Models (LLMs)**, like ChatGPT, to **assist Cybersecurity analysts**.

🔍 2. **Problem Section (Top Left Box)**

Icon: 🔍

Key Points:

- Cybersecurity analysts must handle **large volumes of data**: logs, code, settings, etc.
- **Manual analysis** is not only slow, but also **complex and mentally exhausting**.

👉 This highlights the **need for automation or intelligent assistance**.

🤖 3. **Solution Explored (Top Right Box)**

Icon: 🤖

Key Points:

- LLMs (like ChatGPT) can act as **AI assistants**.
- They can:
  - Summarize complex security data
  - Detect vulnerabilities
  - Interpret system logs

👉 This shows how **AI can reduce analyst workload** while improving insights.

📚 4. **Case Studies (Bottom Left Box)**

Icon: ✔️

Two real-world examples of AI helping analysts:

- **Android Vulnerability Detection**: LLMs analyze Android apps for flaws.
- **Security Log Analysis**: LLMs review logs to detect suspicious behavior.

👉 Demonstrates **practical applications** of LLMs in Cybersecurity.

⚠️ 5. **Challenges (Bottom Right Box)**

Icon: ⚠️

Risks of relying on LLMs:

- They may **hallucinate** or give **inaccurate results**.
- Risk of **leaking sensitive data** if prompts are not handled securely.
- They behave like a **black box** — it's hard to understand *how* they reached a conclusion.

👉 Warns users to **use LLMs carefully**, not blindly trust them.

🎯 Final Message of the Slide:

LLMs are powerful tools that can help in Cybersecurity analysis, **but they must be used responsibly** due to certain risks and limitations.

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